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How artificial intelligence is reshaping information literacy in academic libraries: A global Scientometric analysis (2020–2025)

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ABSTRACT

Artificial intelligence (AI) is consistently reshaping information literacy (IL) library services, training, and professional practices in academic libraries. This study provides a scientometric analysis of scholarly communication to investigate the integration of AI, IL, and academic libraries from 2020 to 2025. To do so, we obtained the bibliographic data from Web of Science, Scopus, ScienceDirect, EBSCO, and SpringerLink. We retained 1669 out of a total of 6500 publications, which were assessed by productivity metrics, global and time-normalized citation metrics, and network analysis of journals, authors, nations, and keywords. The results indicate focused and rapidly growing research published in leading journals. The Journal of Academic Librarianship emerged highly productive journal, publishing 92 articles (5.5%) over the study period, being a central node of journal co-citation networks. Moreover, it indicates how AI is shaping the IL services and align academic libraries as essential venues for the practical application of AI in information literacy services and education, especially in staff capacity building, ethical management, and assisted use of generative AI applications. The country-level analyses show that the United States emerged the best performing country for AI-IL research production and collaboration. Nevertheless, other contributions, such as those from Asia and Africa, indicate that the number of researchers in this field is growing despite structural discrepancies. The research will present the first comprehensive, data-based analysis of trends in publications, conceptual models, and the development of AI-related information literacy in scholarly libraries, providing empirical evidence on the global development of the LIS field.

Introduction

Artificial intelligence (AI) has evolved from a speculative notion to an actual reality. It is changing the process of knowledge creation, acquisition, and application. At first, university libraries were regarded as sources of knowledge, and with the presence of AI, the quality of services has been enhanced. It includes adaptive learning systems, automated metadata generation, and adaptive query and recommendation systems (Arlitsch & Newell, 2017; Cox et al., 2019; Herron, 2017). The connection between AI and information literacy (IL) has become even more important as AI is changing the information environment. Information literacy (IL) has traditionally focused on academic information skills, including how to find, analyze, and apply information from academic sources in scholarly databases, books, and peer-reviewed texts (Shiri, 2024). IL is also needed in AI-based contexts, where one must be critical of the information produced by an AI, learn to recognize bias in intelligent systems, and exercise responsibility with AI tools in academic settings (Cox, 2023; Shiri, 2024). Such AI-inspired technologies as intelligent agents, chatbots, recommender systems, and natural language processing (NLP) interfaces are being explored to enhance IL initiatives and services in libraries. The technologies offer personalised, engaging approaches for learners to develop key skills (Lichtenthaler, 2018). AI has changed libraries, prompting changes to their information literacy programs. At this point, chatbots and intelligent agents help library users with complicated reference questions and provide better assistance than traditional professionals could (Herron, 2017; Meert-Williston & Sandieson, 2019). The responsiveness of IL tools can be improved through NLP, which assists systems to provide the answer for complex or incomplete queries (Nyga et al., 2018). Also, AI helps manage resources better: machine learning is more effective at classifying metadata (Phillips & Chen, 2017), and combining subject headings with tools like Wikipedia makes them more visible (Joorabchi

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& Mahdi, 2018). These strategies, working in the context of information literacy (IL) programmes, supplement and enhance library services (Arlitsch & Newell, 2017; Fernandez, 2016; Massis, 2018). In these frameworks, AI is transforming teaching and learning, offering flexible, interactive, and learner-centred education that goes beyond training in fixed skills.

The shift has initiated a broad scholarly debate over how to rethink IL in the era of artificial intelligence. Yan et al. (2024) applied bibliometric mapping to the knowledge structure of data literacy and showed that quantitative mapping is an effective technique for determining the emergent educational paradigm. By comparison, Hicks et al. (2023) emphasized the interdisciplinary role of IL ideas, integrating them with other disciplines. Likewise, Han (2025) examined the development of topics in AI literacy-related studies, presenting the conceptual progress of this field using bibliometric indicators and a systematic review. These articles highlight the potential of AI to restructure the pedagogical model of IL courses by incorporating algorithmic learning, critical evaluation of intelligent systems, and ethics training into library-centred pedagogical models. These important findings have not yet been fully mapped in the global relationship among AI, IL, and academic libraries. Most of the recent literature has been dominated by professional views of the intelligent library (Cox et al., 2019), conceptual taxonomies (Shiri, 2024), or the broader implications of AI in libraries to society (Arlitsch & Newell, 2017). The following discussion would be enhanced by a critical scientometric lens that tracks the knowledge-generation process in the discipline, defines the leading journals and the most prominent authors, and pinpoints the regional input to this dynamic discussion. Although previously researchers focused on AI and information literacy as a concept or locally, academic librarians need to be provided with evidence-based directions that will guide them in designing instruction, developing staff, and planning the services. The purpose of the study is to support the professional choices of librarians through the systematic mapping of the research patterns in the world, popular journals, and emerging topics at the intersection of AI, information literacy, and academic libraries. In such a manner, it will be able not only to characterize the academic environment but also facilitate the professional decision-making of librarians in the environment of AI-driven change (Cox, 2023; Hervieux & Wheatley, 2021).

Significance of the study

This work addresses a gap using a scientometric method to map international research on the overlap of AI and IL in academic libraries. Between 2020 and 2025, taking into account leading journals, scholars (famous), publications with a high number of references (citation pattern), and publications at the national/regional level. The findings demonstrate the transfer of knowledge across disciplines and geographical locations, as well as the orientation toward productivity and impact. The research will help identify journals, and research centres, clarify emerging themes and gaps, and thus make a significant contribution to academics and practitioners. The results can also be applied in practice to academic libraries, as they can use the findings to plan AI-based IL programs, professional development, and evidence-based decision-making. In addition, the study establishes the significance of academic libraries for the future, placing them in the context of discussions of digital change globally, the ethical application of AI, and equal access to knowledge.

Objectives of the study

This paper addresses the topic of incorporating AI technologies as mediators of information literacy services and instruction in academic libraries, rather than AI literacy as a standalone content area.

1. To examine evidence-based trends in AI-enabled IL services, instruction & research in academic libraries (2020–2025) by an analysis of prominent journals, publishing patterns, and international research contributions.
2. To analyze citation patterns in AI-enabled IL services, instruction & research inside academic libraries from 2020 to 2025 by examining author and journal co-citation networks, globally cited documents, and trending papers through time-normalized global citations.
3. To identify research themes and hubs in AI-enabled IL services, instruction & research (2020–2025) using keywords, authors, and journals within academic libraries.

Research questions

The following are the research questions that will guide this study and outline the bibliometric analyses and visualizations that will be presented in the results section:

RQ1. What has been the trend of research studies on AI integrated IL service, instruction and research in academic libraries from 2020 to 2025 in terms of yearly growth of publications, top journals, and international research contributions?

RQ2. How do the dynamics of citation and intellectual structure of AI and IL research vary between academic libraries between 2020 and 2025, as measured by author and journal co-citation networks, publications with the most citations, and time-normalized global citation indicators?

RQ3. Which prevailing and emergent research themes, thematic clusters, and knowledge hubs are it possible to identify in AI and information literacy research in academic libraries over the period of 2020–2025 through the use of keyword co-occurrence, author output, and journal analysis?

Literature review

AI and IL in academic libraries have remained and expanded in scope in recent decades, becoming the focus of library and information science, education, and technology studies. This section will provide a concise picture of the significant research trends that have shaped the field and prepared the methodology of the present scientometric analysis, rather than a comprehensive systematic appraisal. We conduct a literature review by subject matter to demonstrate the key perspectives, significant problems, and research gaps in the field of AI and IL. This overview also clarifies the study's bibliometric objectives and how our analysis extends previous work by monitoring what is being published, the most influential work, and what is being researched globally.

Artificial intelligence and information literacy in academic libraries

With the emergence of AI, academic libraries and institutions of higher learning have been identified as key points in advancing IL and digital transformation. According to bibliometric analyses, the number of AI-related studies in the field of library and information science (LIS) has grown considerably by 2020, with the focus on the changing role of libraries in the development of digital and information competencies in the age of AI (Wu et al., 2024a, 2024b; Islam & Guangwei, 2025; Kulkanjanapiban et al., 2025). This increasing number of literature demonstrates the rapid adoption of AI technologies by the academic and library environment. According to recent empirical studies, although academic libraries are slowly becoming interested in AI, the degree of preparation, organisational capacity, and even moral readiness vary considerably across the fields and contexts (Ali et al., 2020; Olusegun Oyetola et al., 2023). Research conducted in developing countries shows that intentions to implement policies and professional interest in AI often exceed access to technical expertise, personnel training, and support for long-term infrastructure (Harisanty et al., 2023; Hodonu-Wusu, 2024; Jabeen, 2025). The ethical issues of data privacy, algorithmic

discrimination, and control have become a central theme in discussions of AI use in academic libraries (Hussain, 2023; Inamdar, 2023). Altogether, these results suggest that AI-related studies within the LIS are influenced by the impact of technology and contextual, institutional, and regional factors, and the use of extensive scientometrics analysis is necessary to outline the global research patterns and their effects (Islam & Aziz, 2023; Wu et al., 2024a, 2024b).

AI as a concept of literacy and as a service tool in the library

A large part of LIS research has focused on integrating AI, literacy, and library services. A study on AI literacy reveals a shift away from traditional approaches to IL toward skills such as computational thinking and algorithmic familiarity. It is observed that research on AI literacy is increasingly focusing on the concept of IL, which is more closely correlated with technical and analytical skills (Tenório et al., 2023). On the other hand, Jantakun et al. (2025) highlighted the transformation of the vision of digital literacy in the academic context through AI. Likewise, both Chen et al. (2021) and Gong et al. (2025) described the increasing complexity and interdisciplinary nature of teaching IL at higher education. In addition to such a literacy-centric view, other studies view AI as a technological solution to improve library and educational services. Okunlaya et al. (2022) have developed a theoretical model of AI-based library services that support digital transformation through institutional means. Empirical studies in many different places also demonstrate this tool-oriented approach. Olusegun Oyetola et al. (2023) analyzed the adoption of AI in Nigerian libraries, paying particular attention to infrastructural and preparedness issues. Ali et al. (2020) and Ali et al. (2022) conducted SWOT analyses of the opportunities and constraints of South Asian academic libraries. Jabeen (2025) concludes that Pakistani academic libraries are relatively well equipped with policies and processes regarding AI, but staff skills and training, as well as sufficient resources/infrastructure, are lacking. The research claims that to ensure sustainable adoption of AI, specific capacity-building, enhanced governance, and investment in digital infrastructure will be necessary. Moreover, AI is considered a technology that enhances services and a new form of literacy, implying a gradual shift in the operational use of activities to a broader aspect of adoption in academic libraries within the context of educational and pedagogical activity.

Optimism, uncertainty, and ethical considerations of implementing AI

In addition to services and literacy, the contribution of AI to libraries has been addressed within the broader technological, professional, and educational contexts. Cox (2023) demonstrates how AI is transforming librarians' work and the skills required in academic libraries. AI is also described by Frederick (2023) as the continuation of the history of automation in libraries. Friesen et al. (2023) focus on the application of AI in selected disciplines, whereas broader education studies tend to believe that AI can transform teaching and learning on a large scale (Bahroun et al., 2023; Gibson et al., 2023). According to the literature, there are growing concerns about ethical, professional, and social implications. The application of AI has often been highlighted in recent literature on the topic through the ethical and inclusive implementation (Harisanty et al., 2023; Hodonu-Wusu, 2024; Hussain, 2023; Inamdar, 2023). However, they expressed concerns about the possibility of misuse of AI-generated content in libraries. It has been demonstrated that bibliometric research on AI studies in LIS is increasingly global and collaborative and tends to be associated with big-data methodology in the Fourth Industrial Revolution (4IR) (Islam et al., 2024). Meanwhile, the combination of positive and negative perceptions indicates that AI is not uniformly applied in academic libraries, as institutional resources, staffing, ethical behavior, and local circumstances determine its adoption (Jabeen, 2025).

Although previous bibliometric and scientometric analyses have mapped the research on AI in the field of library and information science, most have treated AI either as a general technological trend in libraries or as a component of digital or AI literacy in education. Current discussions tend to focus on AI adoption, service automation, and national and regional trends, failing to examine information literacy as the core mediating framework within which AI transforms the practice of academic libraries. However, the current study makes information literacy, rather than AI itself, the central analytical perspective and examines how AI technologies converge with IL services, teaching, and professional practice in the context of academic libraries in particular. This work provides a more case-specific and practice-based perspective on how to understand information literacy discourse as it may be changing in the global community through the lenses of publication dynamics, co-citation structures, time-normalized citation impact, and thematic mapping.

Research gap

Although the study of AI in LIS is increasing rapidly, and the connection to IL is becoming increasingly established, the existing literature remains divided in terms of goals and levels of analysis. Most studies focus on AI literacy and competency discussions in the education sector (Jantakun et al., 2025; Gong et al., 2025), but another line of research also explores AI applications, opportunities, and challenges in academic libraries through local or regional case studies (Ali et al., 2020; Harisanty et al., 2023; Jabeen, 2025; Okunlaya et al., 2022; Olusegun Oyetola et al., 2023). Recent bibliometric analyses indicate that AI-related research in LIS is both growing and internationally collaborative (Islam & Aziz, 2023; Islam & Guangwei, 2025; Kulkanjanapiban et al., 2025; Wu et al., 2024a, 2024b).

Nevertheless, even with these contributions, a gap still exists in terms of integrated bibliometric reviews, which analytically study the trends of publications, journals, and documents of influence, trends of international collaboration, citation patterns, and the development of topics at the particular junction of AI, IL, and scholarly libraries. Bibliometric studies on AI in LIS are predominantly conducted on a larger scale or individual indicators, which is restrictive in gathering an all-encompassing comprehension of how the specialised research area has grown globally in recent years. To fill this gap, the given work performs a targeted scientometric analysis of the AI and information literacy studies in academic libraries in the 2020–2025 period and integrates publication dynamics, co-citation network, time-normalized citation impact, and thematic mapping based on keywords. This methodology offers a timely and field-relevant lens of the intellectual framework, emerging research topics, and global knowledge sharing in the form of AI-based information literacy in academic libraries.

Materials and methods

Database selection

The databases were selected due to their broad coverage of papers in the field of LIS. Web of Science and Scopus are frequently employed in bibliometric studies and are widely recognized for their comprehensive indexing (Jabeen, Yun, Rafiq, & Jabeen, 2015; Jabeen, Yun, Rafiq, Jabeen, & Tahir, 2015., Islam & Aziz, 2023). While EBSCO and ScienceDirect deal with discipline-specific and applied research results, Springer offers access to interdisciplinary scholarship. It excluded LISA (ProQuest) due to strong overlap in its core LIS journal coverage with Scopus and Web of Science, which were adequate for retrieving the literature of interest. Moreover, Scopus and Web of Science offer more stable and exportable fields (e.g., cited references, author identifiers, affiliations, funding information, and subject classifications), which are necessary to conduct credible citation, co-citation, and network analysis.

The search in Web of Science and Scopus was conducted in Title,

Abstract, and Author Keywords. The search included the title, abstract, and available keywords fields, which were supported by the searched platforms in ScienceDirect, EBSCO, and SpringerLink. The research did not employ full-text searching to ensure that the retrieved records had a specific focus on AI and information literacy. As the dataset is a mix of citation-indexes databases (Web of Science, Scopus) and publisher-based ones (ScienceDirect, EBSCO, SpringerLink), it is biased toward variation in indexing coverage and citation behavior. The interpretation of citation results is in regard to relative patterns (e.g., visibility and influence) but not the citation counts of databases, which are really equal.

The reliability of scientometric analyses is strengthened and coverage is enhanced when many sources are used (Islam et al., 2024; Jabeen et al., 2017). The final data set, after correction and removal of deduplication, comprised 1669 bibliographic records. Bibliographic records were systematically acquired from five esteemed academic databases: Springer, Web of Science (WoS), ScienceDirect, EBSCO, and Scopus, using a structured Boolean search query. The search term ("Information Literacy" OR "Digital literacy") AND ("AI" OR "Artificial Intelligence") AND ("Libraries" OR "Library & Information Science") was used to find papers that particularly examine the relationship among AI, IL, and academic libraries, see (Fig. 1).

The search string did not include the term 'academic' because many relevant papers do not include 'academic library' in the title, abstract, or keywords, even though the paper is about academic libraries. Including it would have eliminated relevant research. Rather, we used our inclusion/exclusion criteria to guarantee the academic library focus during the screening step.

Validation and refining of search queries

The Boolean search query was developed through repeated refinement to ensure relevance and accuracy. Preliminary pilot searches were conducted using several term combinations relevant to AI, digital literacy, and library contexts. After the initial review, we filtered out words that were not helpful and retained only the phrases that were quite relevant. The last query was investigated to ensure it aligns with the known terms in IL and AI studies, and minor changes were made to enhance precision and recall. The method's repetition improved search accuracy and reduced the risk of missing relevant studies. A search was conducted in March 2025 to ensure that literature indexed between 2020 and 2025 was well covered. After a preliminary search of 6500 records, screening and eligibility checks were conducted based on the relevance of each record to AI, IL, and LIS. DOI matching and title-year comparison have been done to eliminate duplicate documents. The final data set of all the bibliometric studies consisted of 1669 records, including citations, co-authorship, co-citation, and keyword co-occurrence mapping (Table 1).

Once the records from all databases were exported, the results were combined and deduplicated. The screening was done in two phases. Initially, titles and abstracts were checked to eliminate records that were obviously irrelevant. Second, full texts were verified when eligibility could not be determined from the bibliographic information. Two independent reviewers conducted screening, and when there was disagreement, discussions were held to resolve disputes, and, where necessary, a third reviewer was called upon to agree. An exclusion screening log was kept to record the reasons for exclusion.

Database coverage, duplicate information, and bias elimination

The data analyzed were gathered from five major bibliographic databases to enhance coverage and minimize bias from using a single database. The choice of these databases was due to their extensive indexing of high-impact journals, conference proceedings, and book chapters on library and information science, education, and applied research on artificial intelligence. Google Scholar was dropped due to insufficient transparency in its indexing standards, frequent record duplication, and the absence of advanced export and filtering capabilities, which may undermine replicability in extensive bibliometric analysis.

To resolve record overlap across databases, all obtained references were consolidated and methodically deduplicated via DOI matching, and title-year comparisons were subsequently used to eliminate any remaining duplicates. We also added SpringerLink, despite its tendency to introduce some publisher bias, since it helps index conference papers and book chapters that are not indexed by many citation databases. The advantage of this decision is that it broadens the scope of the search. However, it also implies a practical balance between the rigor of the methodology and its simplicity and accessibility.

2) Document Categories: To narrow down the scope and remain relevant, this study only covered specific types of documents, including conference papers, book chapters, review articles, and journal articles, that were published between 2020 and 2025. Such core journals are essential for shaping AI-related studies of IL and library services, as well as for advancing novel concepts and applications. Their inclusion will also help incorporate various recent trends and studies in this field.
3) Quality and 3) Precision: Peer review ensures reliability and credibility, and this process applies to conference papers, review articles, book chapters, and journal articles. Focusing on these sources, the research thoroughly and methodically examined the research results and found key academic contributions to the field of AI in the library settings.
4) Temporal Relevance: The 2020–2025 period was chosen to represent the rapid pace of AI research at important stage of the digital revolution. This timeline offers an illuminating view of how libraries are incorporating AI to address new opportunities and challenges in IL services and programs, as the applications of AI have not been a particular concern in IL until recently.
5) Method: This paper used bibliometric analysis to establish trends and research designs. Bibliometric OR scientometric methods highlight significant research nodes and offer statistical analysis of large datasets (Dai et al., 2020; Wang et al., 2019). Time-normalized



Fig. 1. Flow diagram of the scientometric analysis.

Table 1
Protocol for search record selection of AI and IL research (2020–2025).

Stage	Database / Source	Search string	Search fields	Filters / criteria applied	Records (n)
Identification	Web of Science, Scopus, ScienceDirect, EBSCO, SpringerLink	("Information literacy" OR "Digital literacy") AND ("AI" OR "Artificial Intelligence") AND ("Libraries" OR "Library & Information Science")	Title, Abstract, Author Keywords	Publication years: 2020–2025; Document types: journal articles, review articles, conference papers, book chapters; Language: English	6500
Screening	All databases combined	Same as above	Same as above	Preliminary screening based on scope relevancy; removal of clearly irrelevant records and duplicates at the database level.	5112
Eligibility	All databases combined	–	–	Publications that strongly integrate artificial intelligence-related technology with information literacy in library and information science contexts	1886
Included	Final bibliometric dataset	–	–	Final records saved after data cleansing and deduplication	1669

global citations (TGC/t) were used to reduce the citation-age bias that privileges older publications and to more accurately reflect new and rapidly evolving research subjects in artificial intelligence and information literacy. Due to the rapid development of AI-related research, recent nevertheless significant papers may not have acquired substantial citation counts; normalising citations by publication year enables a better evaluation of academic influence within a brief and dynamic timeframe. Although field-weighted metrics like FWCI and average citations per paper (CPP) are frequently utilised, TGC/t was considered preferable for recognising trending contributions in an interdisciplinary context where citation practices differ across disciplines and time periods (Chen, 2006; van Eck & Waltman, 2010). We employed two prominent scientometric tools, CiteSpace and VOSviewer. CiteSpace is a tool for finding significant events and depicting the evolution of knowledge domains (Chen, 2006). omains (Chen, 2006). Science visualization and co-occurrence network mapping are supported by VOSviewer (van Eck & Waltman, 2010). This study retrieved information from several bibliographic databases, including Web of Science (WoS), ScienceDirect, EBSCO, and Scopus. In order to determine how many times a publication (or author, journal, etc.) has been cited throughout the database, we computed global citation counts (GC) for every source. Additionally, since the current study focuses on impactful AI domains that are growing rather than only historically significant, we computed impact Time-Normalized Global Citations (TGC/t) and volume of publications (PSC), as

$$\frac{\mathrm{TGC}}{t} = \frac{GC}{\mathrm{end}_{\{\mathrm{year}\}} - \mathrm{PY} + 1} \qquad (where\ end_{year} = 2025) \qquad (1)$$

significant contributions were defined by global citation counts (GC) of more than 50. Simultaneously, publications with the highest time-normalized global citations (TGC/t) between 2020 and 2025 were identified as trending papers, indicating growing scholarly influence. Additionally, a bubble plot was used to depict the link between publishing output (PSC) and citation influence (TGC/t). The bubble size was scaled to the total number of citations, making it possible to identify the top journals for the 2020–2025 time-frame.

Results

Publication trends and global research contributions on AI and IL (RQ1)

Publication patterns

A review of the top journals published in 2020–25 indicates that a few core journals heavily influence the literature on AI and IL in the LIS domain. The *Journal of Academic Librarianship* has 92 publications (5.5%), followed by *Communications in Information Literacy* with 65 publications (3.9%) and the *Journal of Information Literacy* with 56 publications (3.4%). Renowned journals like College and Research Libraries and the Journal of Librarianship and Information Science have high levels of representation, indicating their central importance in the field. Specialised journals like the Health Information and Libraries Journal and the Journal of Medical Internet Research are highly effective despite their small numbers, with the latter having an impact factor of 6.0 and an H-index of 214. The publications of the regional journals, the South African Journal of Libraries and Information Science and the Journal of the Australian Library and Information Association, demonstrate that geography and local context are important in AI and information literacy (IL) studies. On the whole, the results indicate that the dominant journals in AI-IL research are those focused on AI. In contrast, a smaller set of interdisciplinary and regional journals contributes to the scholarship selectively yet significantly (see Table 2).

The Journal of Academic Librarianship is the most influential in this area as it is dedicated to higher-education library research and actively publishes the work on professional changes related to AI. This focus indicates an emerging research movement on AI and IL in leading LIS journals, which contributes to the field's initial boundary formation and influences the development of academic discourse (Cox, 2023; Hervieux & Wheatley, 2021).

Trends in the dynamics of cross journals

According to the heat map, publications connecting academic libraries and AI with IL have increased continuously since 2020, with separate spikes in specific journals. The Journal of Academic Librarianship could be considered an active journal, with a peak of 34 in 2023 and remaining at this level through 2024. Further contributions were made by the Journal of Information Literacy, which demonstrated consistent growth, reaching 15 in 2024, and by Communications in Information Literacy, which achieved 27 in 2022.

Individual but significant increases were also observed in the Journal of Medical Internet Research (21 in 2023), the South African Journal of Libraries and Information Science (17 in 2024), and Information Experience and Information Literacy (24 in 2024). Meanwhile, the midrange but continuous publications appeared in journals such as College & Research Libraries, Health Information & Libraries Journal, and Reference Services Review. Such periods of high productivity are depicted by the dark blue cells of the heat-map, which visually emphasize the central events in the area of investigation (Fig. 2). The Global South and regional journals are not very prominent in large research databases. This is usually because they are not properly indexed, they receive fewer citations, and lack international collaboration. Nevertheless, they publish significant local and practical publications on AI and IL, which are usually overlooked in high-impact citation databases (Wu et al., 2024a, 2024b; Harisanty et al., 2025).

Table 2
The 25 best journals associated with ai and information literacy (IL) associated research (2020–25).

Rank	No. of Publications	% of Publications	Journal Name	Country	Impact Factor	H Index
1	92	5.5%	Journal of Academic Librarianship	United States	2.30	72
2	65	3.9%	Communications In Information Literacy	United States	1.10	25
3	56	3.4%	Journal of Information Literacy	United Kingdom	1.51	17
4	43	2.6%	College & Research Libraries	United States	1.70	62
5	42	2.5%	Journal of Librarianship & Information Science	United Kingdom	2.10	41
6	36	2.2%	Library Management	United Kingdom	1.40	41
7	34	2.0%	IFLA Journal	United Kingdom	2.08	29
8	32	1.9%	Journal of Medical Internet Research	Canada	6.00	214
9	29	1.7%	Health Information & Libraries Journal	United Kingdom	0.79	53
10	28	1.7%	Journal of The Australian Library & Information Association	United Kingdom	0.64	27
11	28	1.7%	Library Hi Tech News	United Kingdom	N/A	28
12	28	1.7%	Reference Services Review	United States	1.00	45
13	27	1.6%	Journal of Business & Finance Librarianship	United Kingdom	1.40	21
14	27	1.6%	Portal: Libraries & The Academy	United States	0.90	45
15	26	1.6%	International Information & Library Review	United Kingdom	1.24	32
16	26	1.6%	Journal of Library Administration	United States	0.90	35
17	25	1.5%	South African Journal of Libraries & Information Science	South Africa	0.50	N/A
16	24	1.4%	Information Experience & Information Literacy Springer (CCIS series)	Switzerland	N/A	N/A
19	24	1.4%	Journal of Documentation	United Kingdom	2.00	80
20	23	1.4%	Information Development	United Kingdom	2.10	43

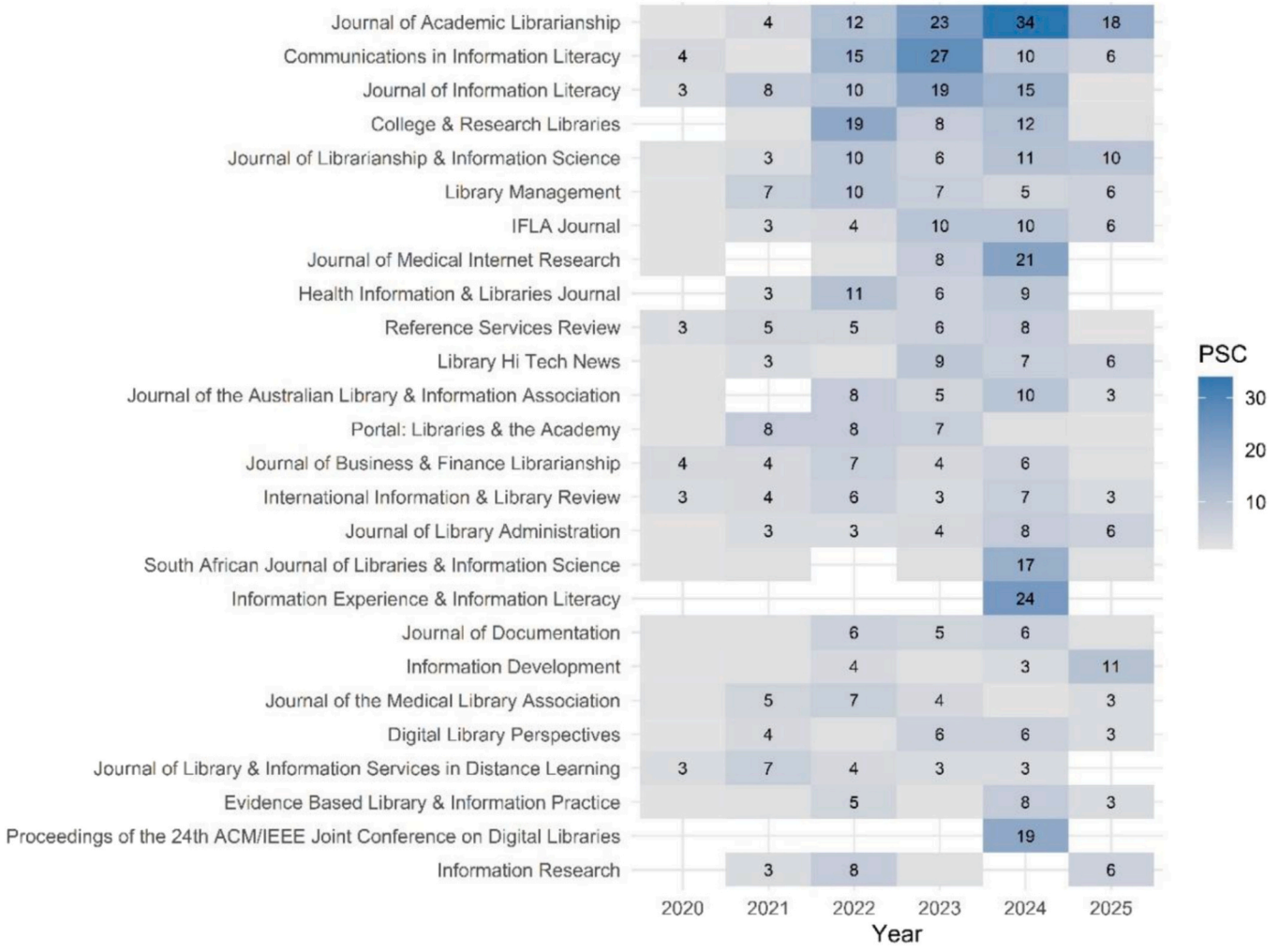


Fig. 2. Heat-map of yearly publications, top 25 journals on AI, Information Literacy, and libraries based on RQ 1.

Global/national diffusion of knowledge

The global collaboration map (2020–2025) shows that the research output is clustered in some countries, especially the top ones. The US is a major centre that produces the majority of publications and is developing strong relationships with partners in Europe, Asia, Africa, and Australia. Other important contributors include China, Australia, Canada, the United Kingdom, and India, which are characterised by high productivity and strong international cooperation. Other emerging countries, including Malaysia, Nigeria, Pakistan, Bangladesh, Ghana and South Africa, have become increasingly prominent, particularly in recent years. The temporal network shows that the number of collaborations has been steadily growing, with 2023–2025 defined by even more growth in developing regions.

Thailand and India, in particular, with institutions such as the OPJ, Mizoram, and Sai Indian universities, are active collaborators with Shinawatra University (Thailand), one of the region's strongest partnerships. Partnerships between South-South partners are observed (e.g., Nigeria-South Africa: 8 papers; Indonesia-Malaysia: 5; Jordan-Malaysia: 4), as well as North-South partnerships (China-USA; Nigeria-USA). Other regional centres include Pakistan, Bangladesh, and Malaysia, where there is a wide range of connecting options, as shown in Fig. 3. The industry is marked by the strong presence of new economies and regionalisation in Asia and Africa, and by global integration centred on the USA. It indicates that AI and IL research is taking shape through a hybridisation of leadership by the global North and increasingly strong South-South networks of cooperative relationships. Fig. 3 shows how these collaborations vary over time. The colour gradient indicates the year of publication: dark red nodes indicate publications from 2025, orange from 2024, yellow from 2023, green from 2022, and blue from 2020. International cooperation has increased over time, and recent years have seen greater participation from Asia and Africa. The results support the dominance of established authorities in research and the uniformity of globalization in scholarly work.

This trend shows that the research infrastructure and funding are unevenly distributed across the world, with such countries as the United States, the United Kingdom, or China acting as coordinators due to their early usage of AI technologies and robust academic publication systems. The increasing participation of Asian and African countries implies further dissemination of AI and IL studies into the developing scholarly context, which signifies a shift to a more diverse and geographically differentiated generation of knowledge (Islam & Aziz, 2023; Wu et al., 2024a, 2024b).

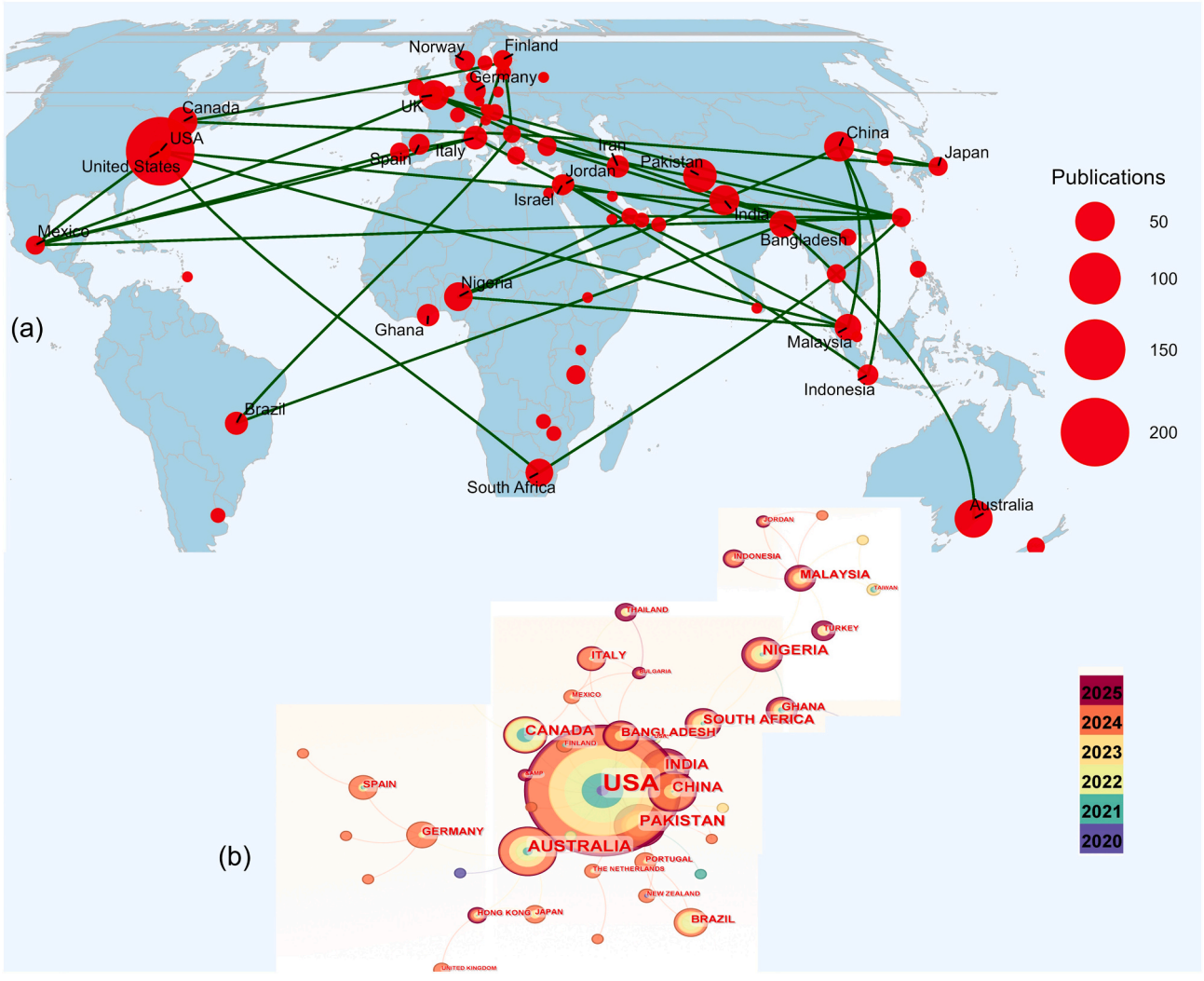


Fig. 3. Top-10 countries in terms of research output in 2020–2025 USA (179), Australia (30), Pakistan (21), China (20), India (19), Nigeria (16), Canada (16), South Africa (13), Bangladesh (12), Malaysia (10) (b) Visualization of the temporal dynamics of collaborations (RQ1).

Co-citation patterns

Dynamics of citation and intellectual structure of AI and IL (RQ2)

Co-citation analysis is a standard bibliometric tool that examines the rate at which two papers are cited in sequence, thus establishing the conceptual map of a research topic (Small, 1973). It is important in scientometric studies because it helps identify groups of authors, journals or documents that share conceptual or theoretical commonality, indicating the knowledge base underlying a given field (White & McCain, 1998). By studying these patterns, scientists can determine how the foundational ideas have evolved, which areas of a field have emerged as research frontiers, and which are core and peripheral within a discipline (Boyack & Klavans, 2010). Co-citation analysis plays a key role in AI and IL research, as it can be used to demonstrate relationships among foundational literature, the conceptualization of important ideas such as AI literacy, and practical models within librarianship, thereby building a unified scholarly field.

Author co-citation network

The three key clusters formed by the co-citation analysis portray different yet similar areas of study in AI and IL. The initial small group, to which researchers like Bingcheng Wang, Davy Tsz Kit Ng, and Dilek Cetindamar belong, is most interested in the recent phenomenon of AI literacy as a measurable and required competence. Their work focuses on establishing the concept, developing assessment tools, and using AI literacy in professional, educational, and personal development contexts. Because it is a specialised and developing field that has not yet been fully integrated into larger discussions in digital literacy, this particular yet creative focus explains the cluster's distinctive positioning away from the mainstream See (Fig. 4(a)). *Allan Martin* and *David Bawden* are representatives of the second tiny cluster, which has its roots in the basic study of computer literacy. Among their contributions are "Origins and Concepts of Digital Literacy" and "A European Framework for Digital Literacy," which provide the theoretical and conceptual foundations for understanding how people interact with digital technologies. This cluster is comparatively isolated from the larger body of research on AI because it reflects an earlier stage in the subject, when digital literacy was defined in more general terms without the pressing influence of AI.

Since they address AI literacy in the particular, formal context of academic libraries, the researchers in the third and largest cluster-among them *Sandy Hervieux, Amanda Wheatley, and Andrew Cox*-attract an impressive amount of co-citation connections, see Fig. 4(a). Their study extends frameworks that connect AI literacy to long-standing traditions in IL, while also exploring how AI is transforming librarianship's professional practices, training, and competencies. This cluster takes up the network as it addresses practical, institution-based problems and links ideas about literacy with the real problems of libraries, educators, and information professionals.

This is supported by fig. 4(b), which indicates that co-citation links are strong in the central cluster. The main contributors to this cluster are Bruce E. Massis, Andrew Cox, Sandy Hervieux, and Amanda Wheatley. Their contributions to the fields of academic libraries, IL education, and professional development are still evident in research. The numerous ties indicate that this is the focal cluster, as it links the theory to the actual institutional dilemmas.

It is evident in Fig. 4(b) that at one point there is a small link: Asefeh Asemi and Jian Wu were co-cited only once, and their work, likely related to computer and digital literacy skills and capabilities, was mentioned. Even such a weak connection is important, as it can indicate a specialised or novice topic that has not yet been well integrated with the mainstream literature. Combined, these two figures indicate that the discipline remains dominated by a significant and well-established body of library-related research. There are also other smaller, more specialised studies on the periphery of the research field, not centrally placed. This is because most prominent scholars in digital literacy are not yet well-networked (thus citing) with researchers interested in the practical application of AI in libraries. Such a weak relationship could merely

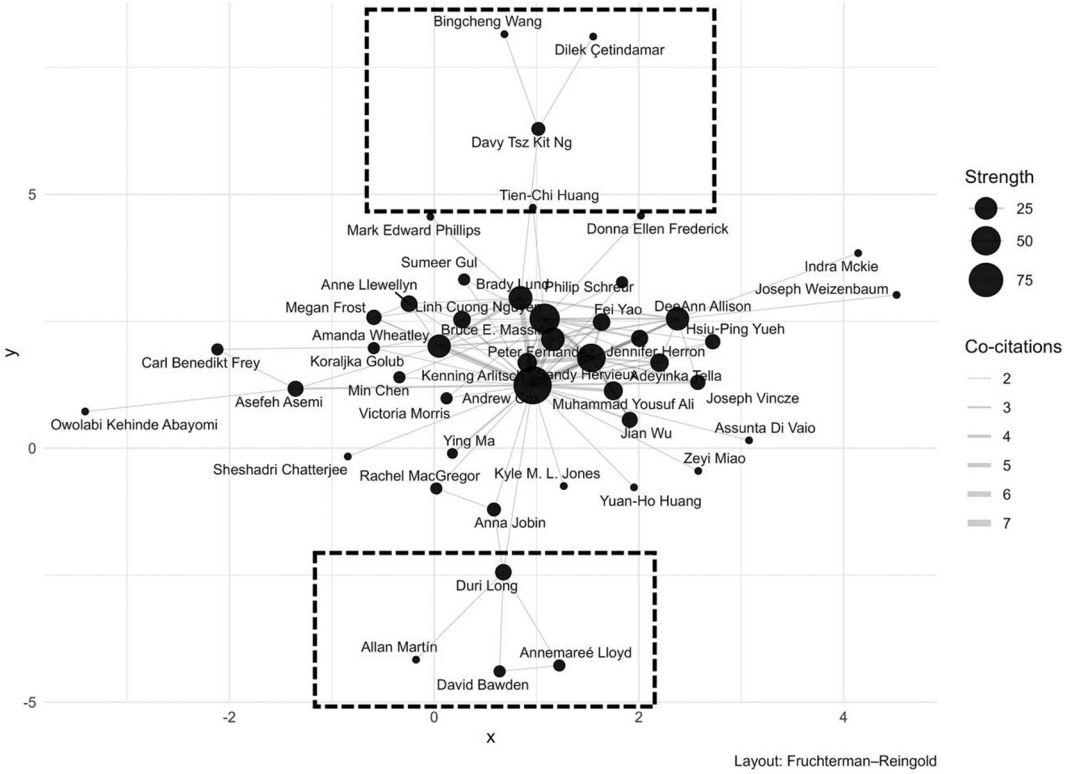


Fig. 4(a). Author co-citation network based on RQ 2.

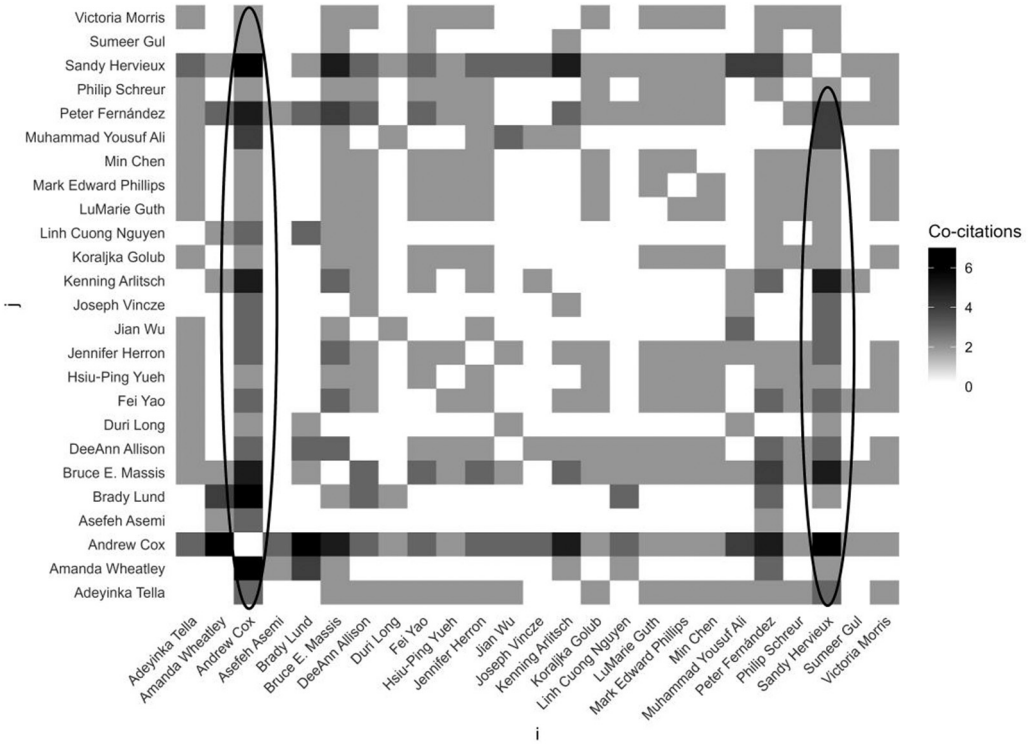


Fig. 4(b). Heat-map of author co-citation network.

indicate a period of transition: AI literacy is in its emergent stages and is already beginning to move beyond the usual IL conceptualizations, though it is not yet a well-established, well-networked field of study (Cox, 2023; Shiri, 2024).

Journal co-citation network

The journal Co-Citation Analysis (JCA) identifies a prominent cluster of 116 journals linked by 444 co-citation links, signifying the core discourse in this study domain. The dominance of library-focused journals is highlighted by the journal citation network, with larger nodes denoting stronger citations. The Journal of Academic Librarianship, The

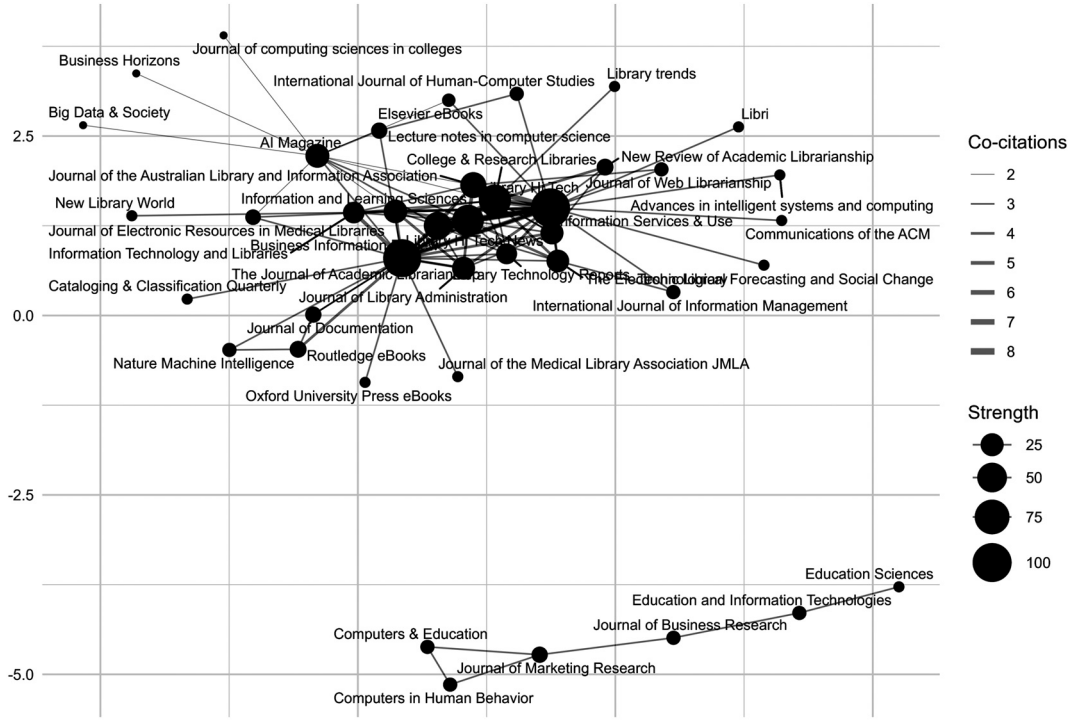


Fig. 5(a). Journals co-citation network (RQ2).

Electronic Library, and College & Research Libraries serve as pivotal centers, connecting specialised and multidisciplinary publications. At the outer edge, two small clusters are visible: one associating education and behavioral science journals (Computers & Education, Education and Information Technologies, Journal of Business Research), and another connecting information science publications (Journal of Information Science, Information Processing & Management). The peripheral categories indicate that the subject of interest, even though most of the study is library-centred, is progressively engaging with closely related fields such as educational technology, user behavior, and information science see Fig. 5(a).

The heat map shows the intensity with which co-citation patterns are observed, with darker cells indicating stronger associations. The most co-citations are to the Journal of Academic Librarianship (8 co-citations), Library Hi Tech (7), and The Electronic Library (7), which can be considered central library journals. These findings are significant because they affect scholarly discourse. Furthermore, there are some practice-focused journals, including Library Technology Reports and Library Hi Tech News, which are also correlated, indicating that a considerable percentage of this literature is focused on practical and professional issues. At the same time, the weak but steady connections with educational and interdisciplinary journals are a sign of the continued expansion of the field's intellectual base. According to the data, there are specific trends between 2020 and 2025. The growing force of digital transformation and AI has made library journals the intellectual backbone of the field, making LIS the most important target of scholarly discourse. At the same time, there is a cross-citation connection among education, behavioral sciences, and information science, which signifies a tendency toward multidisciplinary cooperation. A powerful library-focused centre exists in the research landscape. However, it is also expanding into specialised clusters that denote new trends in digital learning, user behavior, and information management (see Fig. 5(b). (See Fig. 6.)

Most global cited documents

The analysis of publishing patterns (2020–25) indicates that there is still limited research on IL and AI in university libraries. Most of the significant publications in the field are represented by only a few articles. The Journal of Academic Librarianship is the most prolific journal, with high productivity and citation scores, especially in the spheres of AI and IL in academic libraries. The Journal of the Australian Library and Information Association and Library Hi Tech News are other journals that publish fewer articles but achieve higher time-normalized citation rates, suggesting that their selective contributions have a significant impact on the academic community. Mid-tier publications, such as *College & Research Libraries* and the *Journal of Librarianship & Information Science*, maintain consistent production alongside substantial influence. However, high-output journals like Communications in IL generally receive fewer citations in relation to their volume of work. The elevated citation rates of these premier journals signify their prominence as essential LIS publications and the considerable confidence that LIS academics have in them.

Table 2 provides a summary of significant works that address the nexus of IL and AI, with impact normalized for publication year indicated by both gross citation counts (GC) and time-normalized citation scores (TGC). Wang et al. (2023) and Celik (2023) both achieve high normalized citation rates, indicating early acceptance and uptake, and demonstrating the evident trend of AI literacy research dominance. Okunlaya et al. (2022) and Harisanty et al. (2024, 2025) illustrate the effects of conceptual frameworks and systematic reviews on discourse, particularly in Library Hi Tech, a significant platform in the LIS field. The other cluster centres on the digital transformation of libraries with AI. Cox (2023) draws a clear distinction between the adoption of AI and workforce adaptation and professional competencies, demonstrating the connection between technological development and institutional operations. The thematic scope is extended by practical applications, such as regional studies, as seen in Huang (2024) on Taiwan and Lappalainen & Narayanan (2023) on AI-powered library chatbots. Table 3 as a whole

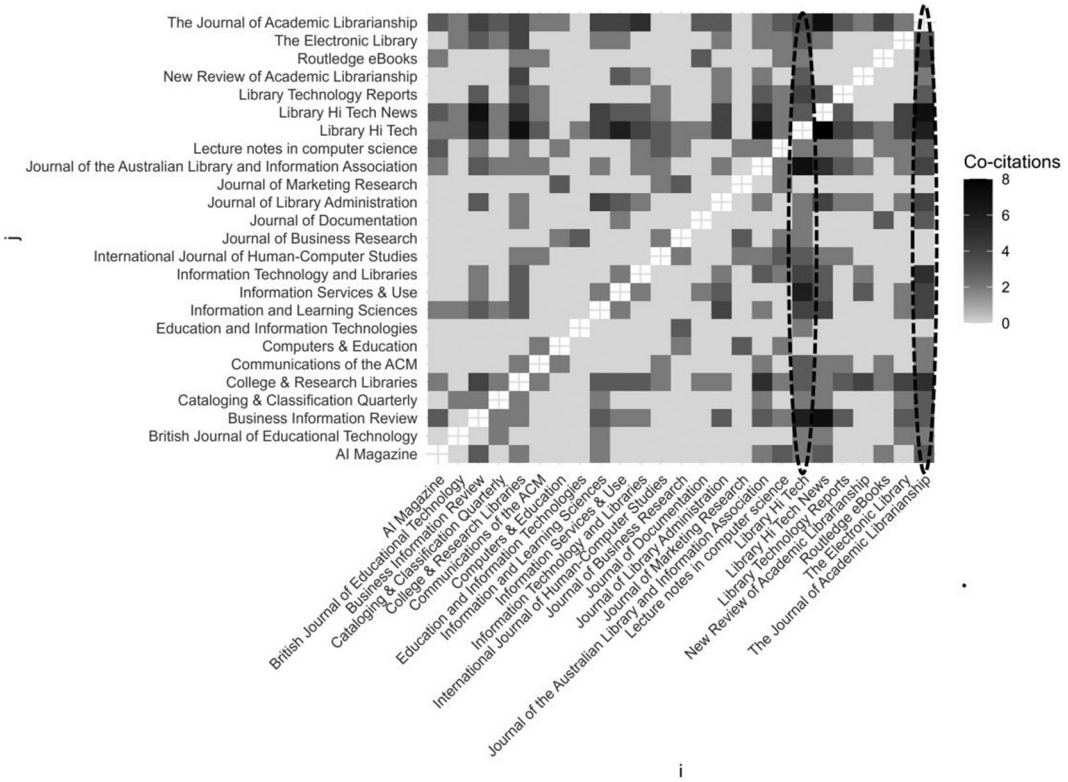


Fig. 5(b). Heat-map of journal co-citations networks.

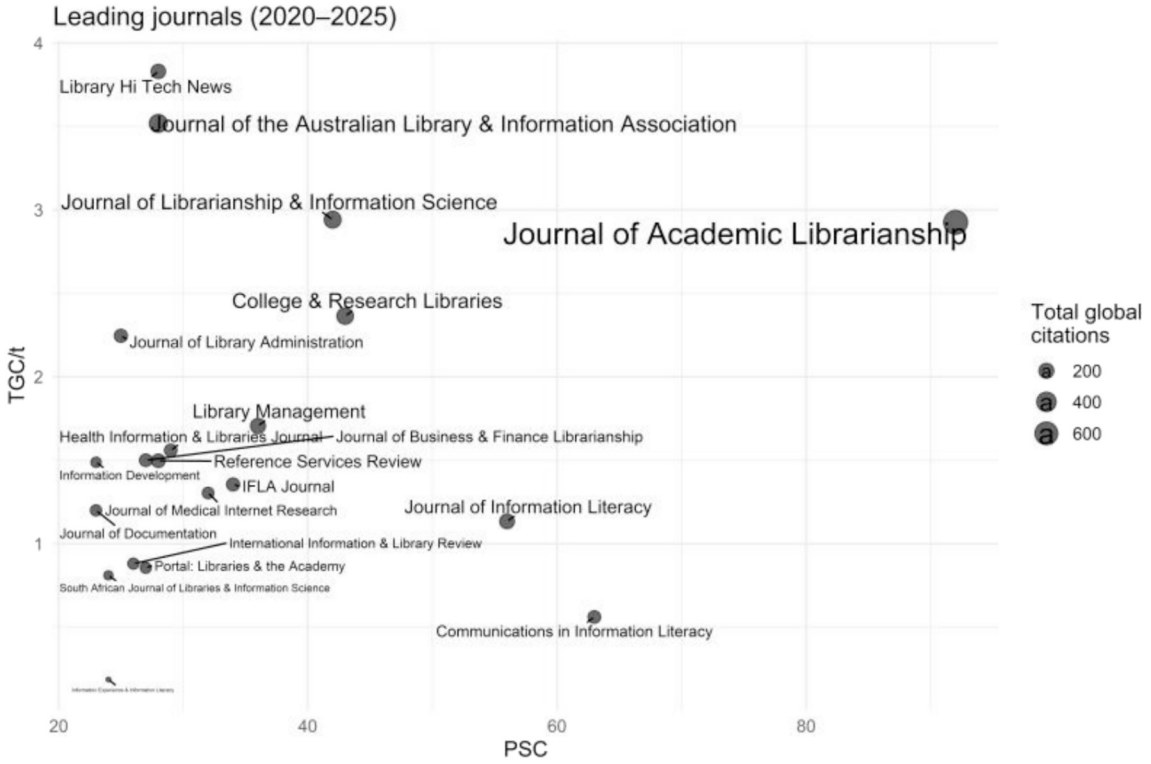


Fig. 6. Top-20 journals by time-normalized global citations (TGC/t) from 2020 to 2025 (RQ 2).

indicates that professional adaptability, digital transformation, and AI literacy are not only the most frequently cited topics but also the ones receiving the most excellent attention from academics.

Keyword co-occurrence network

Mapping research themes and hubs in author–keyword–journal networks (RQ3)

Research Themes and Knowledge Hubs The co-occurrence network in Fig. 7(a) illustrates three unique area clusters. The red cluster comprises the intellectual centre, integrating IL, AI, ChatGPT, chatbots, and library and information science, along with associated phrases like librarians and students. This cluster represents the crossroads between AI and IL, serving as the hub with the most notable links and a significant impact on the rest of the network. The green cluster is more practice-based and encompasses the following terms: academic libraries, digital literacy, curriculum, teaching methodologies, and media literacy, with a focus on the pedagogical and institutional elements of IL activities in higher education. The blue cluster comprises methodological and situational methods, including surveys, questionnaires, descriptive statistics, self-efficacy, and topical issues such as the COVID-19 pandemic, online education, health literacy, and social media. The green and blue clusters bring practical and methodological depth, respectively. In contrast, the red cluster becomes the most influential hub, linking practices and approaches to education across the broader discourse of AI-driven IL (see Fig. 7(a).

Keyword frequency distribution

The frequency distribution in Fig. 7(b) substantiates the dominance of IL, with a highly remarkable occurrence (811 times), and illustrates its essential nature in the field. Libraries (413) and librarians (301) are vital professional and institutional foundations, which confirms the long-term relevance of the profession and the environments. The academic libraries highlight the overall context in which the research was conducted, and AI and the various terms associated with it (AI, ChatGPT, generative AI) demonstrate how technological disruption rapidly impacts IL practices and scholarship Fig. 7(b). Taken together, these standard terms can confirm the critical role of IL in scholarly libraries, with AI serving as a new topical force that is affecting professional roles, teaching methods, and user interactions within the scholarly paradigm of 2020–2025.

Three-way contribution mapping (author–journal–keyword)

The Sankey diagram in LIS is used to highlight the vibrant nature of interactions among authors, research topics, and publication channels. *Alison Hicks*, *Russell Michalak, Muhammad Asif Naveed*, and *Andrew Cox* are among the phenomenological scholars recognized by their broad contributions to several areas and journals. While emergent eras, such as the COVID-19 pandemic and generative AI, demonstrate the field's response to current challenges, thematic analysis reveals IL, academic libraries, and digital literacy as key, well-established areas. Notably, *Russell Michalak* and *Muhammad Asif Naveed* broaden this focus through their work on IL, higher education, and digital competencies. At the same time, *Andrew Cox* and *Alison Hicks* are strongly aligned with digital transformations, contributing to conversations on digital literacy, AI, and generative AI (Fig. 8). With its widespread influence on health IL and library services, the COVID-19 pandemic is associated with several authors, including *Naveed*. It has been published in journals such as the Journal of Academic Librarianship. Collectively, these links demonstrate that although IL remains a dominant and recurring theme, more recent topics, such as digital transformations and pandemic-related literacies, show how LIS studies can be adjusted to meet new technical and global concerns.

Additionally, there is evidence of diverse themes appearing in the data. Authors such as Russell and Michalak discuss AI and higher education, highlighting the expanding impact of pedagogical innovation and digital transformation in the LIS field. Additionally, it is evident

Table 3
Top-10 trending papers (and their authors) by time-normalized global citations (TGC/t).

Title	Author	Journal	GC	TGC/time normalized
Measuring user competence in using artificial intelligence: validity and reliability of artificial intelligence literacy scale.	Wang et al., 2023	Behav. Inf. Technol.	298	99.33
AI library services innovative conceptual framework for the digital transformation of university education	Okunlaya et al., 2022	Libr. Hi Tech	373	93.25
Is adopting artificial intelligence in libraries urgency or a buzzword? A systematic literature review	Harisanty et al., 2025	J. Inf. Sci.	74	74.00
Exploring the Determinants of Artificial Intelligence (AI) Literacy: Digital Divide, Computational Thinking, Cognitive Absorption	Celik, 2023	Telematics & Informatics	220	73.33
Leaders, practitioners, and scientists' awareness of artificial intelligence in libraries: a pilot study	Harisanty et al., 2024	Library Hi Tech	142	71.00
How artificial intelligence might change academic library work: Applying the competencies literature and the theory of the professions	Cox, 2023	J. Assoc. Inf. Sci. Technol.	187	62.33
Exploring the implementation of artificial intelligence applications among academic libraries in Taiwan	Huang, 2024	Library Hi Tech	119	59.50
The CLEAR path: A framework for enhancing Information Literacy through prompt engineering.	Lo, 2023	J. Acad. Librariansh	172	57.33
Mapping the literature on the application of artificial intelligence in libraries (AAIL): a scientometric analysis	Borgohain et al., 2024	Libr. Hi Tech	103	51.50
Aisha: A Custom AI Library Chatbot Using the ChatGPT API	Lappalainen & Narayanan, 2023	J. Web Librariansh	109	36.33

from various regional and contextual contributions, as seen in the work of Batool Syeda Hina, who connect topics such as IL, librarian competencies, and Pakistan to global journals. This tendency to use generative AI as a central term increasingly indicates that the scholarly community is beginning to engage with the new technologies that transform the structure and accessibility of knowledge. Overall, the trend can be described as a two-way process: a single centre of academic librarianship and IL is being supported by new, technology-based, context-specific strands that widen the scope of LIS research.

Discussion

This section presents an interpretation of the bibliometric research results, with reference to the study's research questions and their implications for academic librarianship. This study is a step forward in the discussion of AI in libraries; unlike previous bibliometric studies that map AI research at the general level, this study considers information literacy as the main analytical instrument. The results show that AI is not merely a topic of discussion as a technological implementation, but as a pedagogical and professional issue that is influencing the way librarians teach, direct, and assist learners. The disclosure of the conceptual mapping, new-instruction theme, and ethical focus of this study goes beyond descriptive mapping into practice-based insights on AI in academic librarianship. Regarding library practice, these trends indicate that academic librarians should revise their information literacy instruction, professional education, and ethical practices to promote critical engagement with AI-generated information in teaching and learning settings.

RQ1: Publication dynamics, main sources and international contributions

Research on AI and IL in libraries remains up-and-coming, and most of the published literature is found beyond the Journal Citation Reports (JCR). The trend here indicates that additional research in high-impact, high-reputation journals would be desirable for the region to enhance its academic standing. Simultaneously, conferences like the Information Experience and IL conference proceedings, published in the CCIS series by Springer, offer a more visible global platform, suggesting increased global interest in the topic. Patterns in the publication can further be identified as two distinct trends: some journals, in particular the Journal of Academic Librarianship, have a consistent history of generation over time and stabilise the discussion (Hussain & Ahmad, 2024; Islam & Guangwei, 2025), whereas other sources have temporary spikes in activity that represent the temporary focus. The contributions of regional and developing-country journals (such as the South African Journal of Libraries and Information Science) are increasing, confirming that AI-related IL research is no longer confined to traditional academic centres (Wu et al., 2024a, 2024b). Such minor yet meaningful contributions make the domain of this conversation more inclusive, introducing new viewpoints and better representation of contexts that have been under-researched, helping to augment the growing globalization of this discipline (Harisanty et al., 2025).

These trends demonstrate that the field is in its developing stages. The Journal of Academic Librarianship assists in agenda setting and lending permanency, although it can also focus on what is considered to be core research. Emerging AI topics are being discussed at conferences and non-JCR outlets. The significance of the emergence of regional and Global South journals is that they provide local evidence to a literature that is influenced by the large indexing and publishing systems.

RQ2:Citation structures, intellectual influence, and conceptual shifts

The results of this scientometric analysis show not only the fast increase of the research on the intersection of artificial intelligence and information literacy, but also a definite conceptual change of the way information literacy is interpreted in academic libraries. The fact that keywords including artificial intelligence, AI literacy, ChatGPT, and generative AI are rapidly increasing in use, and that the time-normalized citation impact of AI-literacy research is high, suggests that information literacy is moving toward being conceptualized as an AI-integrated practice and is no longer viewed as a collection of conventional information-seeking skills. This emerging focus on the capacity of users to communicate, access, judge, and critically assess information created by algorithms (Wang et al., 2023; Celik, 2023; Shiri, 2024). Patterns of author and journal co-citation also confirm this change and reveal a close relationship between the existing information literacy research

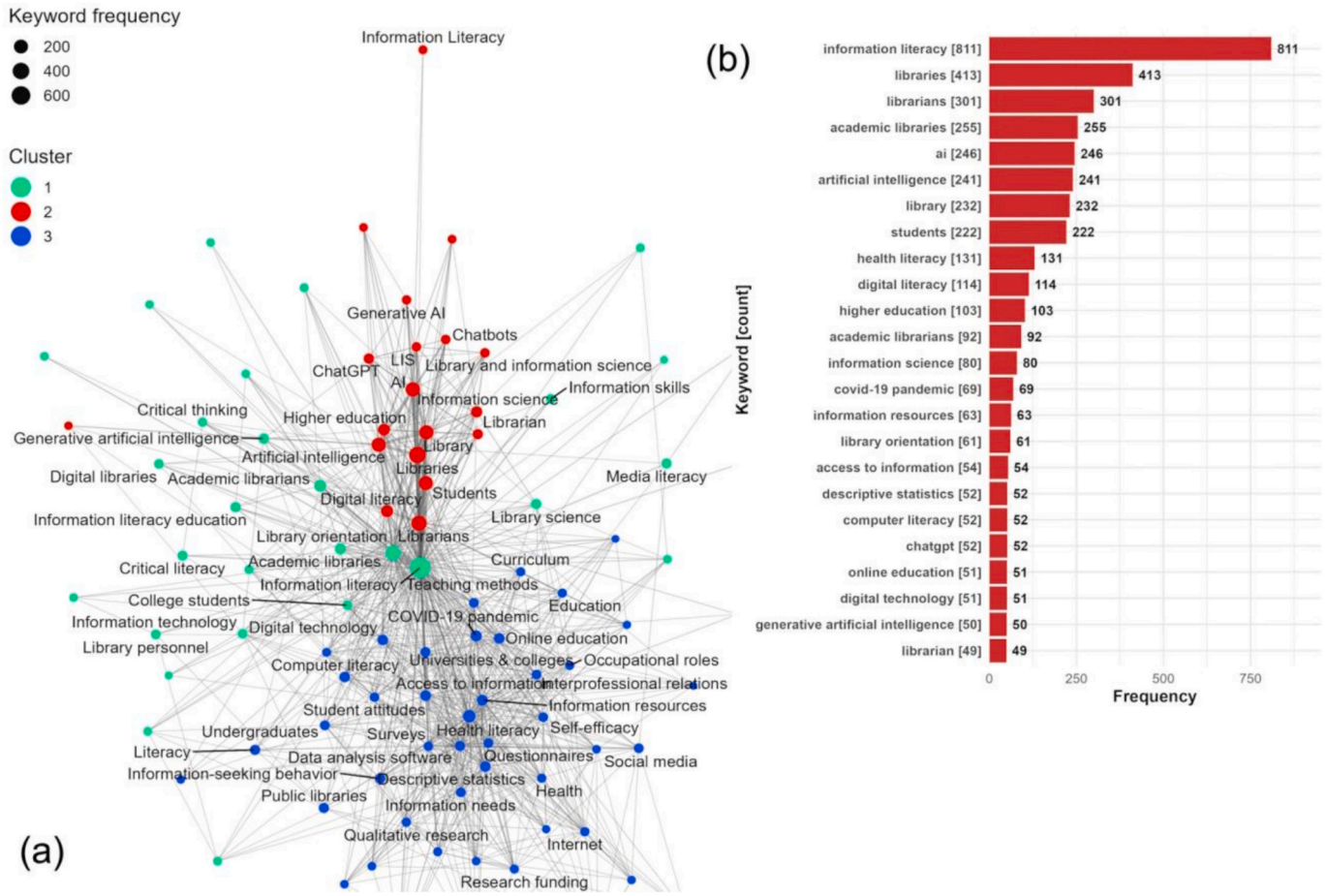


Fig. 7. (a) Keywords co-occurrence network. Node size and colour represents the number of keywords and cluster. Lines of different colours show that the 2 keywords appear in an article. (b) Keywords Frequency Distribution based on RQ3,

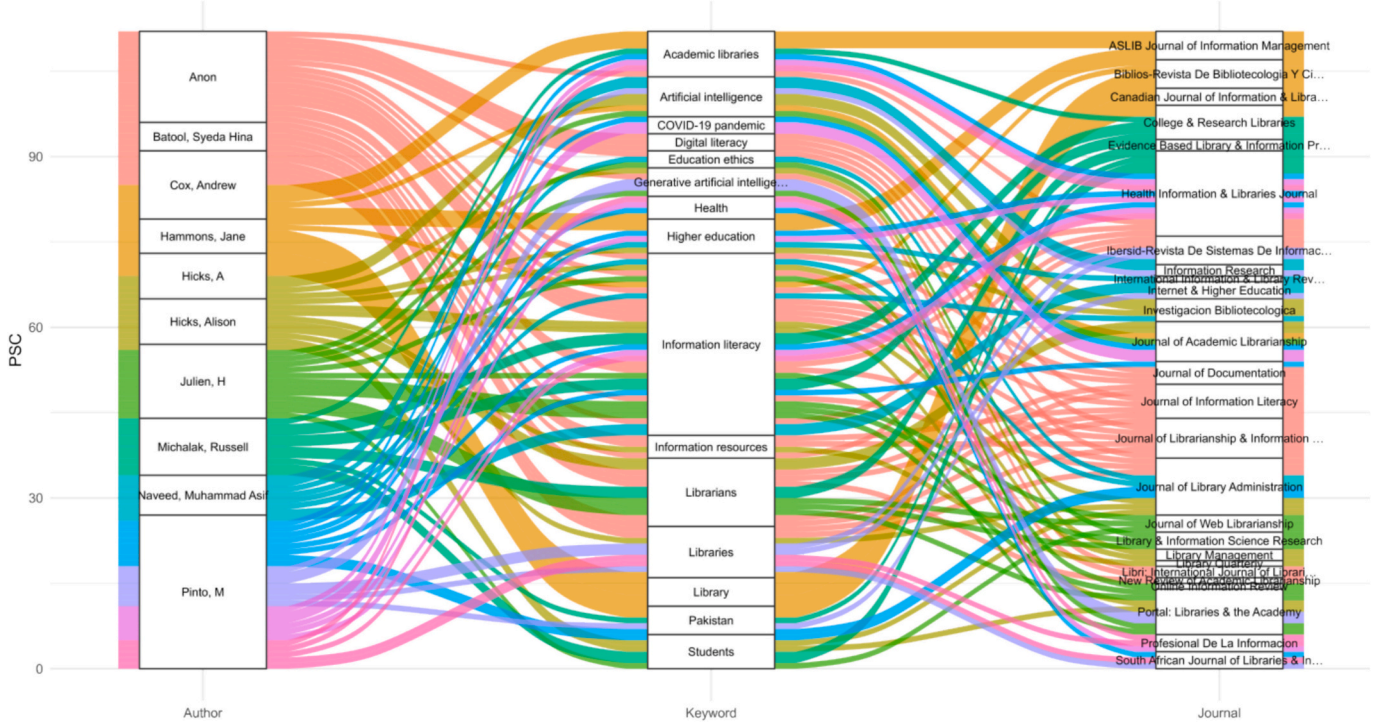


Fig. 8. Top 10 matrix author-keywords-journal contribution (RQ 3).

and recent work on AI competencies, professional roles, and ethical interaction with intelligent systems. The literature is more likely to treat AI as a structural force, redefining the information practices and the instructional role of librarians, instead of situating it as a peripheral tool. Therefore, the scope of information literacy is mandatory to provide awareness of algorithmic bias and responsible usage of generative AI applications and critical analysis of knowledge production, as academic libraries are recognized as significant sites to support user navigation in an AI-integrated information environment (Cox, 2023; Hervieux & Wheatley, 2021).

Recent research indicates that the increased attention to AI does not indicate that it is either extensively or substantially used in library practice. Citation peaks can thus be interpreted not merely as increasing the number of publications, but also as points at which AI has begun to transition from an emerging trend to a common practice in libraries (Harisanty et al., 2025). This change aligns with librarians' perceptions, as more publications indicate greater institutional engagement with AI since 2021, with the highest output between 2022 and 2024 (Yoon et al., 2022). The patterns of publication also indicate that studies of AI and information literacy (IL) are confined to a few journals and are influenced by editorial policies and author groups. The Journal of Academic Librarianship has become one of the most prominent sources through its international outreach, emphasis on higher education, and willingness to embrace innovation in providing library services and professional roles (Hervieux & Wheatley, 2021; Jabeen et al., 2016). On the contrary, Library Hi Tech and Library Hi Tech News focus on current technologies and practical examples, including librarians' willingness to collaborate with robotics and AI-powered systems (Okunlaya et al., 2022; Tella, 2020). However, Communications in Information Literacy provides important critical perspectives on pedagogy; its tendency to be more descriptive limits its broader impact (Špiranec et al., 2019).

Regarding RQ2, the structure of citation and different tendencies indicate that AI literacy has started to create an increasing impact on IL research, and it has not yet been fully absorbed into one, consistent model. Rather, various clusters portay various professional discourses, such as some are dealing with pedagogy, ethics, competencies, or even technology adoption, which shows that the sphere continues to debate what AI literacy entails for librarianship. To academic librarians, this lack of consensus implies that they require flexible information literacy models that can remain aligned with AI, enable students to model AI results, and engage in responsible professional practice. The results suggest that the concept of information literacy that librarianship is converging toward is an AI-informed, but always in flux, one, which is determined not by theory but by practice.

These results indicate that academic librarians need to move beyond AI experimentation and incorporate it into their regular information literacy instruction, reference, and course design. It implies that librarians are expected to assist students and other library users in thoroughly checking AI-generated information, apply generative AI in guided learning activities, and maintain oversight by establishing strict ethical guidelines and local library regulations.

RQ3: Research themes, knowledge hubs, and new global perspectives

Several researchers argue that a significant part of the AI discussion in LIS is not evidence-based but aspirational (Cox, 2023; Cox et al., 2019; Fernandez, 2016). It shows the necessity to match AI expectations with the actual institutional circumstances. This opinion is supported by bibliometric evidence indicating that, despite increased interest in AI and generative technologies, IL remains the core theme of academic library research over the next five years (2020–25) (Wu et al., 2024a, 2024b). In contrast to previous research, which treated AI as a situational variable, the current analysis treats AI as a structural variable closely related to IL, pedagogy, and professional practice, and likely to integrate into the academic setting more quickly and profoundly.

The analysis also shows the growing role of author and journal hubs, which are seen as bridges between conventional literacy structures and technological change. Researchers such as Cox (2023), Hussain and Ahmad (2024), and Michalak and Naveed demonstrate that the concept of classical library competencies is increasingly addressed in the context of AI-based change, a sign that speculation is shifting into early practice. Co-citation patterns also confirm a relationship among library science, education, behavioral science, and information studies, indicating that a multidisciplinary landscape is emerging. Such topics as pandemic-related issues, such as COVID-19 and health literacy, are entrenched in this discourse instead of being short-lived topics. The findings indicate that the links among pandemic literacies, digital access, and user self-efficacy are sustained, which is why this research topic will remain relevant in the study of LIS in the long run, unlike previous beliefs that studies about the crisis would shrink fast (Dai et al., 2020).

Lastly, research output from Asia, Africa, and other parts of the Global South continues to grow, while structural inequalities act as barriers to visibility and influence. The editorial and indexing systems prefer well-established journals of the Global North, which can sideline local or regional research. It would be valuable to deal with these barriers to create a more inclusive and globally representative perception of AI literacy in academic libraries (Wu et al., 2024a, 2024b; Harisanty et al., 2025).

Regarding RQ3, thematic structure indicates an evident transition of speculative description of AI to practice-oriented research where information literacy is at the center stage. The fact that a limited number of authors and journal hubs are prominent is important as the communities are now determining the way AI is translated into library competencies and IL teaching, which is significant since it determines the norms of the future professional presence. The increasing production of the Global South also indicates an increased involvement, but reduced visibility also indicates indexing priorities and the coverage of the databases, rather than a high quality of research. In the case of academic librarianship, these tendencies indicate that context-specific evidence is needed, more IL frameworks of generative AI are required, and the broader scholarly framework of oncology publishing systems should be recognized more fairly.

Limitations

This research has certain limitations. The datasets was based on the sampling databases, and citation patterns and retrieval may be biased by the differences in coverage of indexing and practices. The search terms and bibliographic fields used limit the analysis, so some of the relevant studies that were using other terminologies might not have been captured. Moreover, citation-based indicators provide relative visibility instead of a measure of research quality or practice in academic libraries.

Conclusion

The paper explored information literacy (IL) services, teaching and research trends, and intellectual developments in academic libraries using AI-enabled services during the period 2020–25. RQ1 demonstrated that investigations in this domain have been on the rise but remain confined to a few high-impact LIS journals, with international involvement still uneven. RQ2 revealed that patterns of citation and co-citation indicated a growing but disjointed intellectual framework, suggesting increased interest in AI-enabled IL practices and a lack of a theoretical framework. RQ3 revealed the transition of aspirational and conceptual conversations to practice-driven studies of AI-enabled IL services and instruction, indicated by recognizable author, journal, and thematic hubs, and Global South scholarship is underrepresented.

Between 2020 and 2025, the understanding of AI and IL in academic libraries has been informed primarily by select high-end LIS journals. However, other significant sources have also been interdisciplinary and region-focused. The trends of cooperation indicate a distinct shift from a

Western-dominated literature to greater collaboration with Asian, African, and broader Global South partners, reflecting a more inclusive and interconnected world of research. Both citation and co-citation analysis indicate three fundamental strands of scholarship, namely, known digital literacy theory, AI literacy as an emergent practical skill, and the growing practical use of AI literacy in academic libraries. All these tendencies point to the fact that conceptual debate is replaced by applied engagement. AI literacy is not a side street or an experimental area; it is now becoming part of IL research. Academic libraries are now being presented as locations where professional reskilling, pedagogical innovation, and responsible use of AI can be found. Future studies should thus leave the exploratory debate and turn to evidence-based practices, ethical governance, and capacity building. In this regard, libraries must be actively involved in creating equitable and sustainable AI-driven knowledge spaces.

Finally, this paper identifies three practical priorities for academic librarians as AI is incorporated into information literacy practice. First, libraries are expected to train students to be AI-literate, enabling them to critically assess AI-generated content, identify bias, and use generative tools ethically, fairly, and sustainably. Second, AI systems are to be used as assistant systems that can be controlled by humans, not as independent decision-makers in the context of learning and services. Third, libraries ought to enhance institutional capacity by developing staff and ensuring clear local policies that promote responsible, equitable, and sustainable use of AI.

CRediT authorship contribution statement

Munazza Jabeen: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Claudia Lux:** Validation, Supervision. **Vivien Petras:** Supervision, Resources.

Author statement

All authors contributed substantially to the conception and design of the study, data collection, analysis, and interpretation. The manuscript was drafted with input from all authors, who approved the final version.

Declaration of competing interest

The authors declare that there is no conflict of interest related to this work.

Data availability

Data will be made available on request.

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